Differentiable 3-Axis CNC Machining: Co-Optimizing Toolpaths and Feedrates via Smooth-to-Hard Surrogate Gradients and Residual Reinforcement Learning

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Abstract—Generating physically optimal toolpaths for 3-axis Computer Numerical Control (CNC) milling requires reasoning over long-horizon trajectories in which every cutting pass permanently mutates the in-process workpiece (IPW) geometry and governs subsequent cutter-workpiece engagement (CWE) loads. Classical computer-aided manufacturing (CAM) relies on conservative geometric offset heuristics and constant feedrates because Boolean solid-modeling kernels are discontinuous—yielding zero gradients during air-cutting and undefined derivatives at stock boundaries—while naive geometry-only optimization under exact-stop kinematics fails to exploit real CNC corner blending or respect volumetric chip-load limits. We present DiffCAM, an MIT-licensed, GPU-accelerated dual-mode differentiable 3-axis milling framework that couples exact Boolean swept-cylinder signed distance field (SDF) carving with a log-sum-exp smooth relaxation, Straight-Through Estimator (STE) stiffness continuation, acceleration-bounded junction corner blending aligned with NIST RS274 (G64 P0.05) and Haas Next-Gen Control (G187 P2 E0.05), and an analytical Peak Material Removal Rate (MRR) barrier. Using a drift-free, wall-clearance-weighted PositionSpline co-parameterization of 2.5D XY waypoints and per-segment feedrates, we conduct a controlled study comparing first-order differentiable trajectory and policy optimization (STE-Continuation-Adam, DiffPolicy-SHAC) against zero-order reinforcement learning and black-box search (Hard-PPO, Hard-CEM, Hard-CMA-ES) across 16 canonical Facing, Pocket, Adaptive, and Contour tasks under a strict 64-evaluation budget. First-order STE-Continuation-Adam reduces mean cycle time by $\mathbf{58.5 \pm 6.8}\%$ at $\mathbf{100}\%$ hard geometric and safety feasibility, while first-order DiffPolicy-SHAC achieves nearly $2\times$ the cycle-time reduction of zero-order Hard-PPO ($\mathbf{42.0}\%$ vs. $\mathbf{21.5}\%$) and transfers zero-shot to unseen test parts with a $\mathbf{40.4}\%$ speedup in < 1.5 ms. Ablations show that removing our volumetric MRR barrier induces $+128.3\%$ cutter overload, and independent verification on a 0.125 mm posted G-code evaluator and containerized CAMotics v1.2.0 certifies 8/8 watertight STL solids with 0.00 mm^3 target gouge alongside shop-ready Haas Mini Mill NGC programs.

I. Introduction

Subtractive CNC machining remains the backbone of precision mechanical manufacturing, yet toolpath generation is still dominated by purely geometric rules devised decades ago [1, 7]. Unlike collision-free robot motion planning in static environments, 3-axis milling is a history-dependent, irreversible manipulation process over a continuous volumetric field: every millimeter traveled by the rotating endmill permanently carves the in-process workpiece (IPW), determining the radial and axial cutter-workpiece engagement (CWE) encountered by future passes [12, 18]. Consequently, standard industrial CAM packages decouple toolpath topology (e.g., zig-zag facing, offset pocketing, trochoidal clearing, exterior contouring) from conservative, worst-case constant feedrates chosen to survive full-width slotting corners—leaving substantial machine productivity unrealized during partial-engagement cuts and air-repositioning transitions.

Bridging modern gradient-based trajectory optimization and reinforcement learning (RL) with CNC milling faces three coupled obstacles:

1) Discontinuous Boolean Material Removal: In solid modeling and dexel/voxel kernels [3, 19], subtracting a swept tool volume ψ_{tool} from stock SDF ϕ_t via $\phi_{t+1}(\mathbf{x}) = \max(\phi_t(\mathbf{x}), -\psi_{\text{tool}}(\mathbf{x}))$ yields identically zero spatial gradients when a segment cuts air and non-differentiable kinks at cutter engagement boundaries [16].
2) Surrogate Bias and Cumulative Drift: Naive smooth relaxations of $\max(\cdot, \cdot)$ introduce volumetric occupancy inflation at low stiffness k, causing optimizers to hallucinate phantom material removal that fails exact Boolean evaluation [11, 17]. Simultaneously, parameterizing multi-pass toolpaths via relative segment deltas $\Delta\mathbf{p}_t$ induces $O(T)$ cumulative position variance across $T \sim 50$–150 segments, slamming interior passes into blueprint walls.
3) Controller-Blind Kinematics: Optimizing path length under constant feedrates and exact-stop (G61) assumptions stalls when part tolerances fix the required pass count, whereas naively maximizing feedrates without volumetric engagement feedback causes catastrophic cutter overload ($\text{MRR}_t \gg \text{MRR}_{\max}$) or mismatches physical CNC corner-rounding deceleration [4, 10].

These challenges motivate a central question in differentiable physical AI: when initialized from valid topological CAM seeds under a tight simulation budget, how do first-order analytical gradients through a smoothed-to-

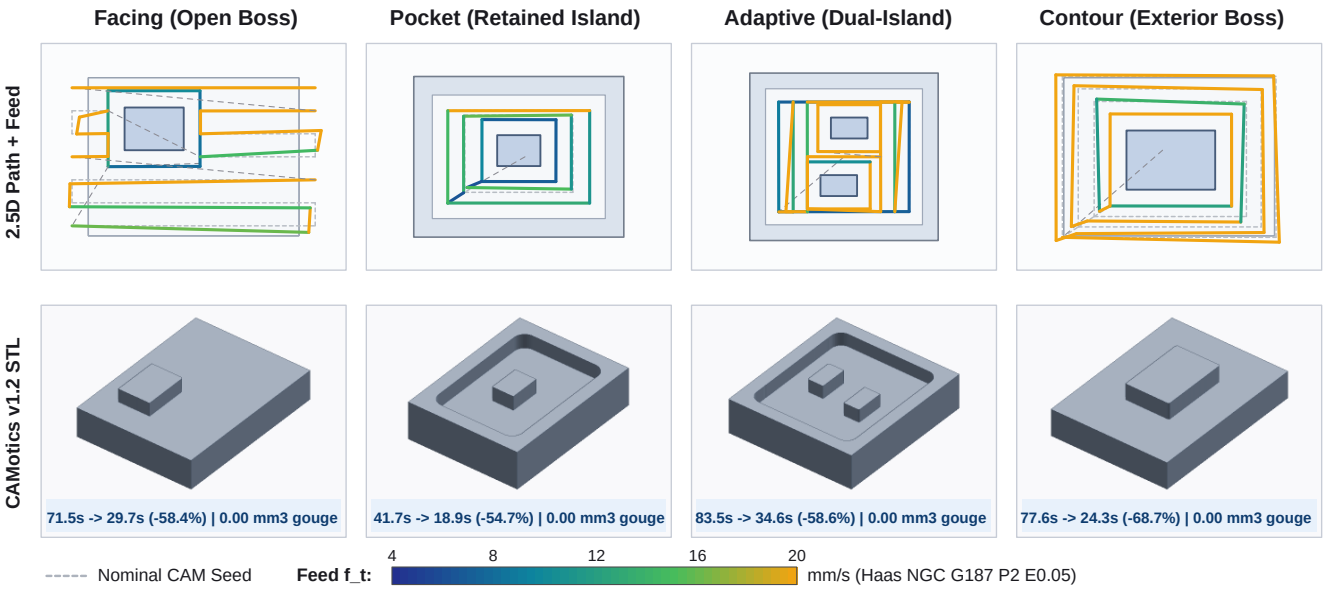


Fig. 1. DiffCAM joint 2.5D toolpath and feedrate co-optimization certified in CAMotics v1.2.0. Top row: Representative held-out test parts across four canonical MIT clean-room CAM families showing stock billets, retained islands/bosses, dashed gray nominal CAM seeds, and optimized STE-Continuation-Adam trajectories colored by per-segment feedrate $f_t \in [4, 20]$ mm/s under Haas Next-Gen Control corner blending (G64 + G187 P2 E0.05). Bottom row: 4×-MSAA isometric renderings of the closed, watertight binary STL meshes reconstructed directly by containerized CAMotics v1.2.0 (0.25 mm grid) from posted Haas NGC G-code, achieving 54.7%–68.7% cycle-time reductions with 0.00 mm^3 target gouge.

hard cutting simulator compare against zero-order RL and evolutionary strategies operating directly on exact Boolean cutting?

To answer this question rigorously, we introduce DiffCAM (Fig. 1), an end-to-end differentiable 3-axis milling simulator, trajectory optimizer, and residual RL benchmark built in Taichi [8]. Our primary contributions are fourfold:

- Dual-Mode Swept-CSG Simulation with STE Continuation: We formulate a GPU-accelerated 3D SDF simulator supporting both exact Boolean material removal and a log-sum-exp smooth surrogate, paired with a Straight-Through Estimator (STE) [2] stiffness schedule ($k : 25 \rightarrow 90$) and exact hard-feasibility checkpointing that eliminates soft-to-hard occupancy drift.
- Blended CNC Kinematics and Volumetric MRR Barrier: We co-optimize continuous 2.5D XY toolpaths and per-segment feedrates under a differentiable acceleration-bounded junction blending model aligned with RS274 G64 P0.05 and Haas Next-Gen Control G187 P2 E0.05, constrained by differentiable pre-cut chip-volume Peak MRR, analytical wall-gouge, holder-collision, and rapid-clearance barriers.
- Drift-Free Co-Parameterization and Controlled RL Study: We introduce an analytical wall-clearance-weighted PositionSpline that eliminates $O(T)$ cumulative delta drift while pinning plunge/retract and finishing boundaries, and compare first-order STE-Continuation-Adam and DiffPolicy-SHAC [20] against zero-order Hard-PPO [15], Hard-CEM [13], and Hard-CMA-ES [6] across a 16-task MIT-licensed benchmark spanning Facing, Pocket, Adaptive, and Contour operations.
- Multi-Fidelity G-Code, CAMotics STL, and Haas NGC Certification: Every held-out test solution is post-processed into standard RS274 and Haas Mini Mill NGC G-code (O01001–O01008) and verified by both a 0.125 mm-step swept-SDF evaluator and containerized CAMotics v1.2.0 [3] (0.25 mm octree), achieving 8/8 watertight STL meshes, 0.00 mm^3 target gouge, and a 59.9% aggregate test cycle-time reduction.

II. Related Work

Geometric CAM and Engagement-Aware Toolpaths. Classical pocket and contour milling rely on 2D Voronoi medial-axis offsets [7] and direction-parallel rastering, which suffer from sharp radial engagement spikes in concave corners. Trochoidal and adaptive clearing strategies [1, 12] bound radial immersion via circular or cycloidal loops, while solid verification engines use Z-buffer dexels [19], R-functions [16], or octree marching-cubes isosurfaces as in CAMotics [3]. However, these forward geometric kernels are non-differentiable, preventing direct gradient-based co-adaptation of waypoint geometry and feed schedules to part-specific stock state.

CNC Corner Blending and Feedrate Scheduling. Physical CNC interpolators avoid coming to a complete stop at every linear G1 block junction (G61) by blending

adjacent velocity vectors within a user-specified path-deviation tolerance—standardized as G64 Pe in NIST RS274NGC [10] and G187 P2 Ee on Haas Next-Gen Control [4, 18]. Classical feedrate scheduling optimizes speed along a frozen reference path under axis acceleration or cutting-force bounds [1, 4]. Conversely, optimizing path geometry under frozen feeds yields negligible speedup when feature walls lock pass topology. DiffCAM unifies both regimes by backpropagating simultaneously through blended junction kinematics, swept chip-volume engagement, and waypoint positions.

Differentiable Physics vs. Zero-Order RL. Differentiable simulators such as DiffTaichi [8] and Brax [5] enable low-variance first-order trajectory optimization and short-horizon analytic actor-critic training (SHAC) [20]. Yet Suh et al. [17] and Metz et al. [11] showed that stiff contact discontinuities can induce high gradient variance or surrogate bias, allowing randomized zero-order estimators such as PPO [14, 15], CEM [13], and CMA-ES [6] to remain competitive. By combining log-sum-exp CSG smoothing with Straight-Through Estimator (STE) evaluation [2] and stiffness continuation, DiffCAM preserves high directional gradient alignment ($\cos\theta \geq 0.98$) while guaranteeing zero surrogate gap at deployment.

III. Differentiable 3-Axis Milling Formulation

A. Dual-Mode Swept-Cylinder SDF Material Removal

We represent the 3D stock billet on a uniform Eulerian grid $\Omega \subset \mathbb{R}^3$ of voxel pitch Δx with signed distance values $\phi_t : \Omega \to \mathbb{R}$ after segment $t \in \{0, \dots, T\}$, following the convention $\phi_t(\mathbf{x}) < 0$ inside solid material and $\phi_t(\mathbf{x}) > 0$ in empty space. Given a flat-end cylindrical cutter of radius R, flute length L_{flute}, holder clearance height L_{clear}, and holder shank radius $R_{\text{holder}} > R$ moving linearly from $\mathbf{p}_{t-1} \in \mathbb{R}^3$ to $\mathbf{p}_t \in \mathbb{R}^3$, let $\tau^\star(\mathbf{x}) = \text{clip}\big(\frac{(\mathbf{x}-\mathbf{p}_{t-1})\cdot\Delta\mathbf{p}_t}{\|\Delta\mathbf{p}_t\|_2^2+\varepsilon}, 0, 1\big)$ denote the closest-point parameter along $\Delta\mathbf{p}_t = \mathbf{p}_t - \mathbf{p}_{t-1}$, with interpolated tool-tip reference $\mathbf{c}_t(\mathbf{x}) = \mathbf{p}_{t-1} + \tau^\star(\mathbf{x})\Delta\mathbf{p}_t$. The local swept cutting-flute SDF is

$$\psi_{\text{tool}}(\mathbf{x}; \mathbf{p}_{t-1}, \mathbf{p}_t) = \max\Big(\|\mathbf{x}^{xy} - \mathbf{c}_t^{xy}(\mathbf{x})\|_2 - R, \; c_t^z(\mathbf{x}) - x^z, \; x^z - \big(c_t^z(\mathbf{x}) + L_{\text{flute}}\big)\Big). \tag{1}$$

In exact hard Boolean mode, material subtraction updates the IPW via the exact CSG difference:

$$\phi_t^{\text{hard}}(\mathbf{x}) = \max\big(\phi_{t-1}^{\text{hard}}(\mathbf{x}), \; -\psi_{\text{tool}}(\mathbf{x}; \mathbf{p}_{t-1}, \mathbf{p}_t)\big). \tag{2}$$

Because $\max(a, b)$ has zero derivative with respect to inactive arguments and a discontinuous subgradient along $a = b$, we pair Eq. (2) with a numerically stabilized smooth log-sum-exp relaxation controlled by inverse-temperature stiffness $k > 0$:

$$\begin{aligned}\phi_t^{(k)}(\mathbf{x}) &= \max_k\big(\phi_{t-1}^{(k)}(\mathbf{x}), -\psi_{\text{tool}}(\mathbf{x})\big) \\ &= m + \frac{1}{k}\log\Big(e^{k(\phi_{t-1}^{(k)}(\mathbf{x})-m)} + e^{k(-\psi_{\text{tool}}(\mathbf{x})-m)}\Big),\end{aligned} \tag{3}$$

where $m = \max(\phi_{t-1}^{(k)}(\mathbf{x}), -\psi_{\text{tool}}(\mathbf{x}))$. Smooth voxel occupancy is mapped via the logistic sigmoid $\sigma_k(\phi) = (1 + e^{k\phi})^{-1} \in (0, 1)$, which converges pointwise to the hard indicator $\mathbb{I}[\phi < 0]$ as $k \to \infty$ while providing dense, analytically differentiable reverse-mode sensitivities $\partial\mathcal{L}_k/\partial\mathbf{p}_t$ across the $\mathcal{O}(1/k)$ boundary layer.

B. Straight-Through Estimator and Stiffness Continuation

Optimizing $\mathcal{L}_k$ at a fixed low stiffness (e.g., $k \leq 20$) suffers from severe surrogate occupancy drift: the $\frac{\log 2}{k}$ additive shift in Eq. (3) accumulates over overlapping passes, inflating soft stock erosion relative to ϕ_T^{hard}. Conversely, initializing directly at $k \geq 100$ concentrates gradients onto a razor-thin shell. We resolve this bias-variance trade-off via a Straight-Through Estimator (STE) with linear stiffness continuation. At optimization step $n \in \{0, \dots, N-1\}$, we schedule $k(n) = k_{\min} + \frac{n}{N-1}(k_{\max} - k_{\min})$ $(25 \to 90)$ and define the forward-hard / backward-smooth objective:

$$\mathcal{L}_{\text{STE}}(\boldsymbol{\theta}; k(n)) = \mathcal{L}_{\text{hard}}(\boldsymbol{\theta}) + \Big(\mathcal{L}_{k(n)}(\boldsymbol{\theta}) - \text{sg}\big[\mathcal{L}_{k(n)}(\boldsymbol{\theta})\big]\Big), \tag{4}$$

where sg$[\cdot]$ is the stop-gradient operator. Every candidate $\boldsymbol{\theta}_n$ is scored and checkpointed strictly on exact Boolean $\mathcal{L}_{\text{hard}}(\boldsymbol{\theta}_n)$ and hard geometric/safety feasibility, while reverse-mode Taichi kernels supply $\nabla_{\boldsymbol{\theta}}\mathcal{L}_{k(n)}(\boldsymbol{\theta}_n)$ at the cost of one additional hard forward pass (< 8 ms on GPU).

C. Blended Acceleration-Bounded CNC Kinematics

To eliminate the mismatch between idealized constant-speed models and modern CNC lookahead controllers, we formulate a differentiable acceleration-bounded corner-blending timing law (BLENDED_ACCEL_BOUNDED) aligned with RS274 G64 P0.05 and Haas NGC G187 P2 E0.05. Let segment t have length $\ell_t = \|\mathbf{p}_t - \mathbf{p}_{t-1}\|_2$, unit tangent $\hat{\mathbf{d}}_t = (\mathbf{p}_t - \mathbf{p}_{t-1})/(\ell_t + \varepsilon)$, and commanded speed $v_t = f_t$ for cutting moves (G1) or $v_t = v_{\text{rapid}}$ for non-cutting rapids (G0). At interior junction t between consecutive cutting moves, the turning angle α_t satisfies $\cos\alpha_t = \hat{\mathbf{d}}_t \cdot \hat{\mathbf{d}}_{t+1}$, yielding the smooth half-angle sine $s_t = \sqrt{\frac{1}{2}(1 - \cos\alpha_t) + \varepsilon}$ and blended junction speed

$$v_{\text{junc},t} = \begin{cases}\min(v_t, v_{t+1})(1 - \kappa_c s_t), & \text{G1–G1 junction,} \\ 0, & \text{otherwise,}\end{cases} \tag{5}$$

with blending factor $\kappa_c = 0.85$ calibrated to a 0.05 mm corner-rounding tolerance under machine acceleration limit $a_{\max}$. With entry speed $v_{\text{in},t} = \min(v_t, v_{\text{junc},t-1})$ and exit speed $v_{\text{out},t} = \min(v_t, v_{\text{junc},t})$ bounded by reachability

over ℓ_t, the segment duration and total cycle time are

$$\tau_t(\mathbf{p},\mathbf{f}) = \frac{\ell_t}{v_t} + \frac{(v_t - v_{\text{in},t})^2 + (v_t - v_{\text{out},t})^2}{2\,a_{\max}\,v_t},$$
$$T_{\text{cycle}} = \sum_{t=1}^{T} \tau_t(\mathbf{p},\mathbf{f}). \tag{6}$$

Crucially, τ_t is jointly differentiable with respect to both waypoint positions $\mathbf{p}$ (via ℓ_t and junction deflection $\cos\alpha_t$) and per-segment feedrates f_t.

D. Volumetric Peak MRR and Machining Safety Barriers

Increasing feedrate f_t reduces τ_t in Eq. (6), but unchecked acceleration in full-width slotting or corner plunges exceeds cutter flute evacuation and spindle torque limits [1]. At each cutting step t, our simulator integrates the pre-cut swept chip volume directly from the evolving IPW field ϕ_{t-1}:

$$\Delta V_t = \Delta x^3 \sum_{\mathbf{x}\in\Omega} \sigma_k\big(-\phi_{t-1}(\mathbf{x})\big)\,\sigma_k\big(-\psi_{\text{tool}}(\mathbf{x};\mathbf{p}_{t-1},\mathbf{p}_t)\big),$$
$$\text{MRR}_t = \frac{\Delta V_t}{\tau_t}. \tag{7}$$

We enforce a rated volumetric removal limit $\text{MRR}_{\max}$ via a horizon-normalized quadratic overload barrier $\mathcal{B}_{\text{MRR}} = w_{\text{MRR}} \sum_{t=1}^{T} \max\big(0, \frac{\text{MRR}_t}{\text{MRR}_{\max}} - 1\big)^2$. Under this coupling, air-cutting and light radial-skimming passes ($\Delta V_t \approx 0$) receive pure cycle-time gradients that push $f_t \to f_{\max}$, whereas heavy roughing passes automatically throttle f_t or shift waypoints laterally to shave ΔV_t.

Given target blueprint SDF $\phi_{\text{target}}(\mathbf{x})$, final geometric accuracy combines overcut gouge $V_{\text{gouge}} = \Delta x^3 \sum_{\mathbf{x}} \mathbb{I}[\phi_{\text{target}}(\mathbf{x}) < 0]\,\sigma_k(\phi_T(\mathbf{x}))$ and uncut residual $V_{\text{res}} = \Delta x^3 \sum_{\mathbf{x}} \mathbb{I}[\phi_{\text{target}}(\mathbf{x}) > 0]\,\sigma_k(-\phi_T(\mathbf{x}))$, augmented by three analytical machine-protection barriers: (1) an exact continuous 2D point-to-segment wall-clearance barrier $\mathcal{B}_{\text{wall}}$ preventing sub-voxel finishing-wall nicks, (2) a cylindrical tool-holder collision barrier $\mathcal{B}_{\text{holder}}$ over $z \in [c_t^z + L_{\text{clear}}, c_t^z + L_{\text{clear}} + L_{\text{holder}}]$ of radius R_{holder} against ϕ_{t-1}, and (3) a G0 rapid-stock intersection barrier $\mathcal{B}_{\text{rapid}}$ guaranteeing zero rapid contact with unmachined stock. The composite loss is

$$\mathcal{L} = w_T \frac{T_{\text{cycle}}}{T_{\text{nom}}} + w_g \bar{V}_{\text{gouge}}^2 + w_r \bar{V}_{\text{res}}^2 + \mathcal{B}_{\text{MRR}} + \mathcal{B}_{\text{wall}} + \mathcal{B}_{\text{holder}} + \mathcal{B}_{\text{rapid}}, \tag{8}$$

with candidate hard feasibility requiring simultaneously zero holder collision, zero rapid stock contact, and target gouge/residual below strict volumetric tolerances.

IV. Trajectory and Policy Co-Optimization

A. Drift-Free Waypoint-Position and Feed Co-Parameterization

A critical failure mode in long-horizon machining optimization is cumulative delta drift: if an optimizer perturbs segment displacements $\Delta\mathbf{p}_t = \Delta\bar{\mathbf{p}}_t + \delta\mathbf{p}_t$, waypoint $\mathbf{p}_t = \mathbf{p}_0 + \sum_{j=1}^{t} \Delta\mathbf{p}_j$ accumulates $O(T)$ random-walk variance across $T \in [35, 130]$ segments, causing downstream finishing passes or vertical retracts to collide with pocket walls even under tiny step sizes.

We eliminate cumulative drift via a bounded Waypoint-Position and Log-Feed Co-Parameterization (JointPathFeedSpline). Given nominal CAM waypoints $\{\bar{\mathbf{p}}_i\}_{i=0}^{T}$ and nominal cutting feeds $\{\bar{f}_t\}_{t=1}^{T}$, we hold axial heights $\bar{p}_i^z$ fixed to preserve constant-layer floor flatness and safe G0 clearance planes, and parameterize horizontal waypoints and segment feeds as

$$\mathbf{p}_i^{xy}(\boldsymbol{\theta}) = \bar{\mathbf{p}}_i^{xy} + r_{xy}\, w_i \tanh\big((\mathbf{B}\boldsymbol{\theta}_{\text{pos}})_i\big),$$
$$f_t(\boldsymbol{\theta}) = \bar{f}_t \exp\big(\rho_f \tanh(\theta_{\text{feed},t})\big), \tag{9}$$

where $\mathbf{B} \in \mathbb{R}^{(T+1)\times K_p}$ is a clamped cubic B-spline basis ($K_p \le 40$), $r_{xy} = 1.8$ mm bounds maximum lateral excursion, and $\rho_f = 0.80$ spans feed multipliers $f_t/\bar{f}_t \in [0.45, 2.22]$. Crucially, $w_i \in [0,1]$ is an analytical blueprint wall-clearance weight:

$$w_i = \begin{cases} 0, & \text{if } i \text{ is a plunge/retract,} \\ \text{clip}\big(\frac{d_{\text{wall}}(\bar{\mathbf{p}}_i) - \delta_{\text{pin}}}{r_{xy}},\, 0,\, 1\big), & \text{otherwise,} \end{cases} \tag{10}$$

where $d_{\text{wall}}(\bar{\mathbf{p}}_i)$ is the exact Euclidean clearance from the cutter periphery at $\bar{\mathbf{p}}_i$ to any retained pocket wall or island boss ($\delta_{\text{pin}} = 0.08$ mm). Equation (10) automatically pins vertical plunge/retract axes and tangent wall-finishing passes ($w_i = 0$) while granting full $O(1)$-bounded lateral mobility to interior roughing passes.

B. Invariant Segment State and Residual RL Policies

To test whether machining heuristics can generalize across part dimensions without re-solving from scratch, we cast segment-level refinement as a finite-horizon Markov Decision Process with an invariant 18-D per-segment state $\mathbf{s}_t \in \mathbb{R}^{18}$ extracted from a single nominal pass: normalized start/end coordinates in stock-box coordinates (6), unit motion direction and normalized length (4), normalized nominal feed and adjacent junction cosines $(\cos\alpha_{t-1}, \cos\alpha_t)$ (3), binary cutting/plunge/retract flags (2), normalized pre-cut chip volume ΔV_t, nominal $\text{MRR}_t/\text{MRR}_{\max}$, local target SDF clearance, and endpoint mobility $(w_{t-1} + w_t)/2$ (3).

A shared two-layer tanh MLP policy $\boldsymbol{\pi}_\psi(\mathbf{a}_t \mid \mathbf{s}_t) = \mathcal{N}(\boldsymbol{\mu}_\psi(\mathbf{s}_t), \text{diag}(\boldsymbol{\sigma}^2))$ (64–64 hidden units, zero-initialized output layer so $\boldsymbol{\mu}_{\psi_0}(\mathbf{s}_t) \equiv \mathbf{0}$ starts exactly at the feasible nominal CAM seed) outputs residual waypoint and log-feed actions $\mathbf{a}_t \in [-1,1]^3$ mapped through Eq. (9) via DirectSegmentActionMap. We compare two controlled policy-learning paradigms sharing identical network architectures and simulator budgets:

1) DiffPolicy-SHAC (First-Order Differentiable RL): Following Short-Horizon Actor-Critic [20], we roll out deterministic/exploratory policy actions $\mathbf{u} = \boldsymbol{\mu}_\psi(\mathbf{S})$ through our STE continuation evaluator (Eq. (4)) and backpropagate analytical simulator

gradients $\nabla_{\mathbf{u}}\mathcal{L}_{k(n)}$ directly into ψ via the chain rule $\nabla_{\psi}\mathcal{L} = (\partial\boldsymbol{\mu}_{\psi}/\partial\psi)^{\top}\mathbf{P}\nabla_{\mathbf{u}}\mathcal{L}_{k(n)}$. Here $\mathbf{P}$ applies decoupled RMS preconditioning across spatial XY and log-feed coordinates so stiff wall-barrier gradients never drown out smooth feedrate sensitivities during Adam [9] updates.

2) Hard-PPO (Zero-Order Clipped Policy Gradient): Operating purely on exact hard Boolean rollouts ($\nabla_{\mathbf{u}}\mathcal{L}$ unavailable), we train $\boldsymbol{\pi}_{\psi}$ alongside a 64–64 ValueCritic $V_{\omega}(\mathbf{s}_t)$ using Clipped Proximal Policy Optimization ($\epsilon = 0.2$) [15] and Generalized Advantage Estimation ($\gamma = 0.99, \lambda = 0.95$) [14], driven by dense per-segment hard rewards $r_t = -\left(\frac{\tau_t}{T_{\text{nom}}} + w_{\text{MRR}} \max(0, \frac{\text{MRR}_t}{\text{MRR}_{\max}} - 1)^2\right) - \frac{1}{T}\mathcal{C}^{\text{hard}}_{\text{terminal}}$.

V. Experimental Setup

MIT Clean-Room 16-Task Milling Suite. We construct a 100% MIT-licensed benchmark suite spanning four canonical 2.5D CAM strategy families—facing_open_boss, pocket_island, adaptive_channel (dual-island trochoidal/channel clearing), and contour_profile (exterior stepped boss)—with 2 development (dev) and 2 held-out test (test) geometric instances per family (16 tasks total, 35–125 motion segments per program). All tasks model Al 6061-T6 billets (30–38 × 24–32 × 8 mm) machined by a 5.0 mm diameter ($R = 2.5$ mm) flat carbide endmill (12 mm flute, 14 mm holder diameter, 8000 RPM, $v_{\text{rapid}} = 60$ mm/s, $a_{\max} = 1200$ mm/s^2, and family-rated $\text{MRR}_{\max} \in [95, 130]$ mm^3/s).

Methods and Compute Parity. Every optimizer is audited by a deterministic ComputeLedger capped at $B = 64$ simulator candidate evaluations across 3 random seeds $(0, 1, 2)$ on an NVIDIA RTX 5070 Ti GPU (0.50 mm optimization grid): (0) Nominal-CAM, (1) Naive-Smooth-Adam (fixed $k = 25$, no STE), (2) Annealed-Adam (k : $20 \to 85$, final hard check only), (3) STE-Continuation-Adam (k : $25 \to 90$ with per-step hard STE selection), (4) DiffPolicy-SHAC (1st-order RL), (5) Hard-PPO (0th-order RL, 8 episodes/batch), (6) Hard-CEM (population 12, elite 25%), and (7) Hard-CMA-ES [6] ($\sigma_0 = 0.22$, population 12).

Independent Posted G-Code and CAMotics v1.2.0 Audit. All held-out test trajectories are post-processed into both NIST RS274 (G64 P0.05) and Haas Mini Mill Next-Gen Control (G54 G64 + G187 P2 E0.05, programs O01001–O01008) and verified by two independent external evaluators at 2× finer spatial resolution (0.25 mm grid): (1) a standalone NC parser and swept-SDF posted_evaluator stepping at 0.125 mm stations (0.00 mm^3 target gouge tolerance, $\leq$ 2.0 mm^3 residual tolerance, full holder and rapid-move safety screen), and (2) containerized CAMotics v1.2.0 (camsim at 0.25 mm octree resolution), auditing closed manifold watertightness and volume consistency on exported binary STL meshes.

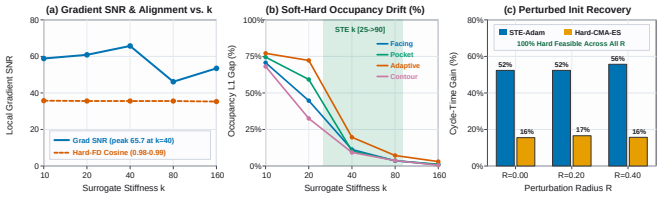


Fig. 2. Exp1: Surrogate gradient conditioning and perturbed recovery. (a) Analytical gradient SNR peaks at $k = 40$ (65.7) while hard finite-difference cosine stays 0.98–0.99. (b) Soft-to-hard occupancy L_1 bias collapses from 72.6% ($k = 10$) to $<4.5\%$ inside our STE continuation window $k \in [25, 90]$. (c) From perturbed initializations $R \in \{0.00, 0.20, 0.40\}$, STE-Continuation-Adam recovers 100% feasibility and 52%–56% speedups vs. 16%–17% for Hard-CMA-ES.

VI. Results and Analysis

A. Exp1: Surrogate Conditioning and Perturbed Reachability

Figure 2 characterizes the numerical mechanics of our smooth CSG relaxation across stiffness $k \in \{10, 20, 40, 80, 160\}$ on the four development families. In Fig. 2(a), directional cosine similarity between analytical smooth gradients $\nabla_{\boldsymbol{\theta}}\mathcal{L}_k$ and central finite differences of exact $\mathcal{L}_{\text{hard}}$ remains remarkably high across all stiffnesses (**0.980**–**0.993**), while local gradient signal-to-noise ratio (SNR) peaks at **65.72** near $k = 40$ before declining to 46.13–53.46 at $k \in \{80, 160\}$ as active boundary voxels thin out. Simultaneously, Fig. 2(b) reveals why naive smooth optimization fails without continuation: the L_1 volumetric occupancy gap between soft and hard IPW states is **72.6**% at $k = 10$ and **52.2**% at $k = 20$, but drops exponentially to **12.6**% at $k = 40$, **4.5**% at $k = 80$, and **1.4**% at $k = 160$. Sweeping k : $25 \to 90$ under STE captures wide-basin SNR early and sharp geometric fidelity late.

In Fig. 2(c), we perturb initial parameters uniformly within $\|\boldsymbol{\theta}_0\|_{\infty} \leq R$ for $R \in \{0.00, 0.20, 0.40\}$ (48-evaluation budget). STE-Continuation-Adam recovers 100% hard feasibility across all R and achieves **52.3**%, **52.4**%, and **55.7**% cycle-time reductions—more than 3.3× the gain of zero-order Hard-CMA-ES (15.5%, 16.6%, 15.7%).

B. Exp2: Single-Task Trajectory and RL Benchmark

Table I and Fig. 3 report our main 16-task × 8-method benchmark (64 simulator evaluations, 3 seeds). STE-Continuation-Adam achieves the strongest performance across every CAM family, cutting mean cycle time from 68.44 s to **27.96** s ($-$**58.5** $\pm$ **6.8**% reduction) at **100**% hard feasibility in 3.40 s of wall-clock GPU time. Comparing smooth gradient baselines isolates the exact contribution of STE: Naive-Smooth-Adam (fixed $k = 25$) stalls at -32.3% because uncalibrated occupancy bias misleads interior roughing passes on Pocket (-22.1%) and Adaptive (-23.2%) tasks, while Annealed-Adam (smooth k-schedule without per-step hard STE checkpointing) reaches -52.8% overall but suffers high variance on

Adaptive clearing ($-41.1 \pm 15.9\%$ vs. $\mathbf{-59.1 \pm 1.1}\%$ with STE).

Comparing our controlled RL arms directly addresses the first- vs. zero-order policy learning question: under identical 64–64 MLP architectures and a 64-episode simulator budget, first-order DiffPolicy-SHAC achieves $\mathbf{-42.0 \pm 19.2}\%$ mean cycle-time reduction (38.18 s), nearly doubling the speedup of zero-order Hard-PPO ($\mathbf{-21.5 \pm 8.1}\%$, 53.67 s) and Hard-CMA-ES (-21.8%). Because Hard-PPO must estimate credit assignment across 40–120 sequential segments from noisy Gaussian action perturbations, 64 rollouts provide only 8 policy updates. By contrast, DiffPolicy-SHAC receives exact per-segment analytical sensitivities on every rollout, converging within 16 evaluations on exterior Contour (-61.7%) and Open Boss Facing (-51.3%). Interestingly, on enclosed pocket_island tasks (-18.4% single-task SHAC vs. -52.9% STE-Continuation-Adam), shared state-conditioned MLP weights couple inner roughing rings with outer wall rings, whereas per-trajectory B-spline parameters adjust individual passes independently.

C. Exp3: Multi-Task Policy Transfer and Physical Ablations

While per-part trajectory optimization excels given 64 evaluations, neural policies amortize optimization across geometries. In Exp3 (Table II top, Fig. 4a), we train multi-task DiffPolicy-SHAC and Hard-PPO policies for 20 epochs across the 8 dev tasks and evaluate transfer on the 8 unseen test tasks:

- Sub-Millisecond Zero-Shot Transfer: In a single deterministic forward pass (<1.5 ms MLP inference, 1 simulator verification), DiffPolicy-SHAC-ZeroShot achieves $\mathbf{-40.4 \pm 6.5}\%$ cycle-time reduction at 100% feasibility—including $\mathbf{-34.8}\%$ on held-out Pocket tasks (surpassing single-task SHAC because multi-task training regularizes ring-transition features). While Hard-PPO-ZeroShot learns a competitive coarse feed heuristic (-39.8%) over 160 multi-task training episodes, its lack of analytical MRR gradients yields $\mathbf{125.9}$ $\mathbf{mm^3}$/s mean peak MRR (+11.1 mm^3/s higher cutter load than SHAC's 114.8 mm^3/s).
- 16-Evaluation Few-Shot Adaptation: Given 16 test-time evaluations, warm-started DiffPolicy-SHAC-FewShot16 improves monotonically to $\mathbf{-43.4 \pm 14.0}\%$ (111.5 mm^3/s peak MRR), approaching from-scratch STE-Adam-16 (-45.7%), whereas Hard-PPO-FewShot16 degrades to -34.6% because stochastic exploration perturbs converged weights over only 4 mini-batch updates.

Our 5-arm ablation on the 8 held-out test tasks (Table II bottom, Fig. 4b) confirms three core physical hypotheses: (1) Path-Only-Frozen-Feed yields only $\mathbf{1.2 \pm 0.9}\%$ speedup because blueprint pocket/island boundaries lock topological pass counts when feedrates cannot adapt to local engagement; (2) co-optimizing XY waypoints and feeds under controller-matched corner blending (Full-Joint-Blended-STE, $\mathbf{58.4}\%$) outperforms both Feed-Only-Frozen-Path (57.8%) and optimizing under an Exact-Stop-Surrogate (57.1%) by relaxing interior corner angles while accelerating low-immersion passes; and (3) setting $w_{\mathrm{MRR}} = 0$ (No-MRR-Barrier) inflates apparent cycle-time gain to 67.0% by commanding maximum feed through full-width slotting cuts, driving mean peak MRR to $\mathbf{254.3}$ $\mathbf{mm^3}$/s ($+\mathbf{128.3}\%$ above rated cutter limits).

D. Exp4: High-Resolution NC, CAMotics STL, and Haas NGC Audit

To verify that optimization on a 0.50 mm grid introduces zero discretization artifacts or post-processing regressions, Table III audits all 8 held-out test parts compiled into Haas Mini Mill Next-Gen Control programs (O01001–O01008, G54 G64 + G187 P2 E0.05). Across all 8/8 test parts, aggregate cycle time drops from $\mathbf{71.03}$ s to $\mathbf{28.47}$ s ($\mathbf{-59.9}\%$). Every program passes our 0.25 mm-voxel, 0.125 mm-step posted_evaluator geometric and holder/rapid safety checks with exactly $\mathbf{0.00}$ $\mathbf{mm^3}$ target gouge and 0.00–1.75 mm^3 uncut residual ($\leq 0.035\%$ of billet volume). Independently, containerized CAMotics v1.2.0 reconstructs $\mathbf{8/8}$ closed, manifold, watertight STL solids (72,752–129,476 facets) whose enclosed volumes match nominal CAM cuts within $<0.15\%$ and exhibit sub-voxel marching-cubes surface deviation (≤ 0.66 mm^3) strictly lower than or equal to Nominal-CAM (≤ 1.75 mm^3).

VII. Limitations and Conclusion

Limitations. First, our current parameterization locks axial coordinates $\bar{p}_i^z$ to canonical 2.5D cutting layers; extending wall-clearance mobility weights to simultaneous 3-axis ball-endmill sculptured surfacing and 5-axis tool-orientation smoothing is a natural next step. Second, while our volumetric Peak MRR barrier prevents steady-state chip-load and torque overload, it does not model frequency-domain regenerative chatter lobes [1] or thin-wall deflection. Third, although DiffPolicy-SHAC enables sub-millisecond zero-shot transfer (-40.4%), topology-changing pass insertion or deletion still relies on upstream geometric CAM seeding. Finally, while our Haas Mini Mill NGC shop packets (O01001–O01008) are certified in CAMotics v1.2.0 and high-resolution swept-SDF simulation, instrumented physical Al 6061-T6 cutting trials measuring spindle load and profilometry roughness constitute our immediate hardware milestone.

Conclusion. We presented DiffCAM, an MIT-licensed dual-mode differentiable 3-axis milling simulator and co-optimization framework that reconciles discontinuous Boolean CSG geometry with smooth analytical policy and trajectory gradients. By combining Straight-Through Estimator stiffness continuation, controller-matched corner-blending kinematics, volumetric Peak MRR barriers, and drift-free PositionSpline parameterizations, DiffCAM

TABLE I

Exp2 Main Benchmark across 16 Milling Tasks (4 Families × 4 Geometries × 3 Seeds, Budget $B = 64$ Simulator Evaluations). Cycle-time reductions (ΔT_{cycle}, %) are mean ± std relative to Nominal-CAM; all methods are evaluated strictly on exact Boolean hard cutting under blended acceleration kinematics.

Method	Optimization Class	Facing (%)	Pocket (%)	Adaptive (%)	Contour (%)	Overall $-\Delta T$ (%)	Mean T (s)	Peak MRR (mm^3/s)
Nominal-CAM	Canonical CAM Seed	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	68.44	66.2
Hard-CEM [13]	0th-Order Elite Search	15.8 ± 2.4	11.8 ± 2.9	9.2 ± 3.7	15.7 ± 2.0	13.1 ± 4.0	59.48	85.9
Hard-PPO [15]	0th-Order Clipped RL	17.8 ± 6.0	18.1 ± 5.9	18.8 ± 4.2	31.2 ± 6.9	21.5 ± 8.1	53.67	92.7
Hard-CMA-ES [6]	0th-Order Covariance ES	22.1 ± 3.0	18.9 ± 6.9	14.8 ± 3.4	31.2 ± 2.6	21.8 ± 7.4	53.62	93.7
Naive-Smooth-Adam	1st-Order Fixed $k = 25$	34.6 ± 6.8	22.1 ± 5.7	23.2 ± 6.6	49.5 ± 4.2	32.3 ± 12.5	45.94	103.8
DiffPolicy-SHAC [20]	1st-Order Analytic RL	51.3 ± 3.6	18.4 ± 14.3	36.5 ± 13.5	61.7 ± 3.0	$\mathbf{42.0 \pm 19.2}$	38.18	107.2
Annealed-Adam	1st-Order Smooth k-Sched	$\mathbf{55.5 \pm 2.6}$	49.6 ± 4.6	41.1 ± 15.9	64.8 ± 2.4	52.8 ± 12.1	32.38	105.3
STE-Continuation-Adam	1st-Order STE Traj (Ours)	55.4 ± 7.5	$\mathbf{52.9 \pm 4.2}$	$\mathbf{59.1 \pm 1.1}$	$\mathbf{66.5 \pm 2.5}$	$\mathbf{58.5 \pm 6.8}$	**27.96**	109.6

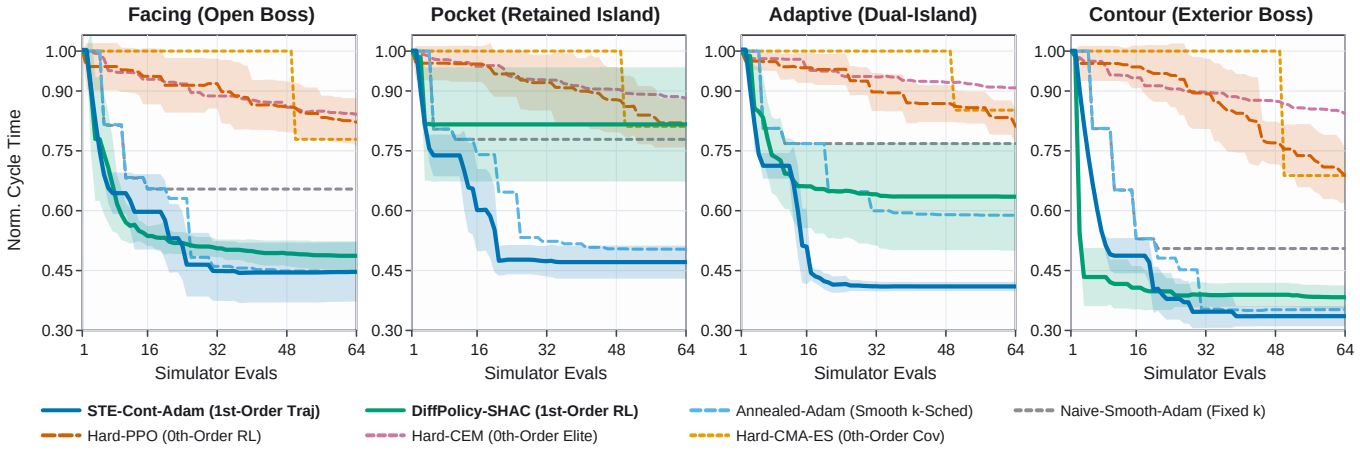


Fig. 3. Exp2 sample-efficiency convergence across canonical CAM families ($B = 64$ simulator evaluations, mean ± 1 SD shaded). First-order methods (STE-Continuation-Adam, DiffPolicy-SHAC) achieve steep cycle-time reductions within the first 16–24 evaluations by exploiting exact per-segment $\partial T_{\text{cycle}}/\partial f_t$ and $\partial \Delta V_t/\partial \mathbf{p}_t$ sensitivities, whereas zero-order RL (Hard-PPO) and evolutionary search (Hard-CMA-ES, Hard-CEM) converge slowly in 100+ parameter dimensions.

TABLE II

Top: Exp3 Multi-Task Policy Transfer (8 Dev → 8 Unseen Test Tasks, 3 Seeds). Bottom: Exp2 5-Arm Physical & Kinematic Ablations (8 Test Tasks, $B = 64$). All arms achieve 100% geometric feasibility; removing the MRR barrier causes severe +128.3% cutter overload.

Regime / Ablation Arm	Evals	$-\Delta T_{\text{cycle}}$ (%)	Peak MRR	Overload
Exp3: Unseen Test Policy Generalization				
Hard-PPO-ZeroShot	1	39.8 ± 2.8	125.9	13.5%
DiffPolicy-SHAC-ZeroShot	1	$\mathbf{40.4 \pm 6.5}$	**114.8**	**4.1%**
Hard-PPO-FewShot16	16	34.6 ± 12.9	108.9	2.2%
DiffPolicy-SHAC-FewShot16	16	43.4 ± 14.0	111.5	3.1%
Scratch-STE-Adam-16	16	$\mathbf{45.7 \pm 5.5}$	**106.3**	**1.4%**
Exp2-Abl: Physical & Kinematic Ablations ($B = 64$)				
Path-Only-Frozen-Feed	64	1.2 ± 0.9	73.8	0.0%
Exact-Stop-Surrogate	64	57.1 ± 6.7	112.5	3.2%
Feed-Only-Frozen-Path	64	57.8 ± 4.9	114.8	3.9%
Full-Joint-Blended-STE	64	$\mathbf{58.4 \pm 6.1}$	115.7	4.8%
No-MRR-Barrier ($w_{\text{MRR}} = 0$)	64	67.0 ± 5.5	**254.3**	**128.3%**

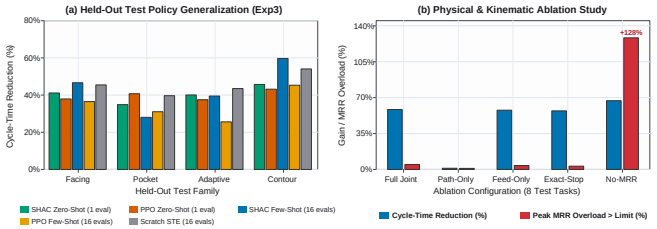


Fig. 4. Exp3 policy generalization and physical ablations. (a) Held-out test cycle-time reductions across families for zero-shot (1 eval) and few-shot (16 evals) transfer. (b) Ablating feed optimization (Path-Only) collapses gains to 1.2%, whereas ablating the volumetric MRR barrier (No-MRR) triggers +128% peak cutter overload.

achieves 58.5% mean cycle-time reductions at 100% feasibility, demonstrates a 2× sample-efficiency advantage for first-order differentiable RL over zero-order RL, and delivers certified watertight CAMotics STL parts and shop-ready Haas Mini Mill NGC G-code.

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TABLE III
Exp4 Independent High-Resolution Verification and Haas Mini Mill NGC Shop Packet Summary Across All 8 Held-Out Test Parts. Audited on posted Haas NGC G-code (G54 G64 + G187 P2 E0.05) via both a 0.125 mm-step swept-SDF posted_evaluator (0.25 mm voxels) and containerized CAMotics v1.2.0 (0.25 mm octree watertight STL reconstruction).

Haas Prog.	Case ID	CAM Family	Stock (mm)	T_{nom} (s)	T_{DiffCAM} (s)	$-\Delta T$ (%)	Gouge (mm^3)	Res. (mm^3)	Safety	CAMotics STL	STL Vol. (mm^3)
O01001	face-test-00	Facing Boss	34×24×8	75.04	32.83	56.3%	0.00	0.00	Pass	Watertight (75.0k△)	4246.1
O01002	face-test-01	Facing Boss	32×24×8	71.50	29.72	58.4%	0.00	0.00	Pass	Watertight (72.8k△)	4371.3
O01003	pocket-test-00	Pocket Island	32×28×8	41.84	21.51	48.6%	0.00	1.75	Pass	Watertight (106.4k△)	5042.5
O01004	pocket-test-01	Pocket Island	34×26×8	41.69	18.88	54.7%	0.00	1.13	Pass	Watertight (102.0k△)	5520.0
O01005	adapt-test-00	Adaptive Dual	38×30×8	98.24	39.25	60.0%	0.00	1.31	Pass	Watertight (129.5k△)	6681.5
O01006	adapt-test-01	Adaptive Dual	36×32×8	83.52	34.57	58.6%	0.00	1.31	Pass	Watertight (127.3k△)	7067.5
O01007	contour-test-00	Contour Boss	32×26×8	78.77	26.67	66.1%	0.00	0.00	Pass	Watertight (76.2k△)	4143.2
O01008	contour-test-01	Contour Boss	32×24×8	77.61	24.29	68.7%	0.00	0.00	Pass	Watertight (72.8k△)	4204.2
Mean / Pass	8 Held-Out Test Parts			**71.03**	**28.47**	**59.9**%	**0.00**	$\leq$**1.75**	**8/8**	**8/8** Watertight	$\Delta V < 0.15\%$

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